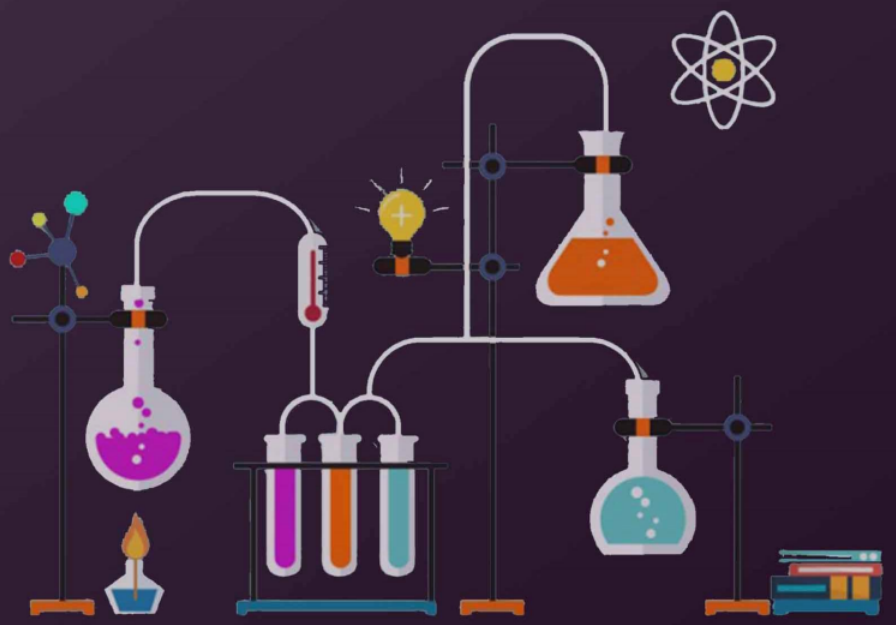
CHEMISTRY

States of Matter



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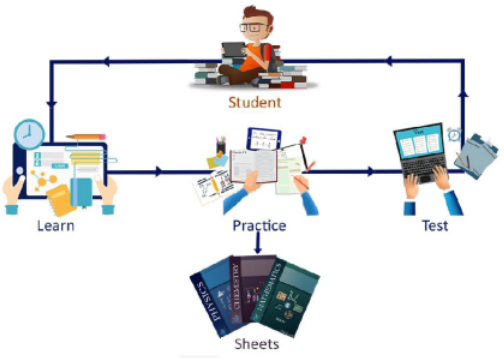
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction	1-2	48		1
2	Parameters of a gas	3-4	1,2,40	1,2	
3	Ideal Gas Laws	5-13	3,4	3	2,30
4	Ideal Gas Equations - Part 1	14-21	49		
5	STP, NTP and Partial Pressure of a Gas - Part 2	22-25	5,6,50	4,5	3
6	Problems on Partial Pressure	26-29	7,8,51	6	4,5,14-17
7	Graham's law of Diffusion	30-33	9-11,41,46, 47,52	7-10	6,18-20,32,33
8	Problems on Piston and SPM		12,30-32	11,12	21-23
9	Kinetic Theory of Gases	34-44	13-17,53,54	13-19	7,27
10	Maxwell's Distribution of Molecular Speeds		18-20,55		8,34
11	Collision Theory	45		20	
12	Real Gas, Van der Waals equation	46-55	21-23,42,56	21-24	9,10,28
13	Real Gas, Compressibility Factor	56-59	24,25,33-39,43,44	25	11,31,35
14	Liquification of gases	60-61	26,27,45	26	12
15	Liquid State	62		27,28	
16	ACC	63	28,29		13,24-26,29

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction

Q1. Which of the following is true about gaseous state :-
(A) Thermal energy = Molecular attraction
(B) Thermal energy >> Molecular attraction
(C) Thermal energy << Molecular attraction
(D) Molecular forces >> Those in liquids

Q2. Molecular interactions between molecules is in order:
(A) Solid<liquid<gas
(B) Solid<gas<liquid
(C) Gas<liquid<solid
(D) Liquid<solid<gas

Parameters of a Gas

Q3. The value of the universal gas constant R depends upon the
(A) Nature of the gas
(B) Mass of the gas
(C) Temperature of the gas
(D) The units of measurement

Q4. Which of the following are correct statements.
(1) 760 torr is equal to 1 atmosphere
(2) 10^6 *dynes/cm*^{2} is called 1 bar
(3) 10^5 *Newtons / m*^{2} is Pascal
(4) Latmosphere is 1.013×10^5*dynes/m*^{2}
(A) 1,3 (B) 1,2
(C) 1,4 (D) 3,4

Ideal Gas Laws

Q5. Temperature below 0 K is not possible. It can be explained by :-
(A) From boyle's law
(B) From avagadro law
(C) From charles law
(D) All of these

Q6. To which of the following gaseous mixtures is Dalton's law not applicable :-
(A) $Ne + He + SO_2$
(B) $NH_3 + HCl + HBr$
(C) $O_2 + N_2 + CO_2$
(D) $N_2 + H_2 + O_2$

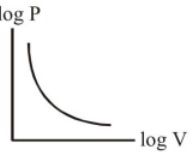
Q7. 5 moles of a gas in a closed vessel was heated from 300K to 600K. The pressure of the gas doubled. The no. of moles of gas will be :-
(A) 5 (B) 2.5
(C) 10 (D) 20

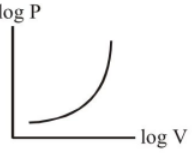
Q8. At 0°C and one atm pressure, a gas occupies 100 cc. If the pressure is increased to one and a half-time and temperature is increased by one-third of absolute temperature, then final volume of the gas will be :-
(A) 80 cc (B) 88.9 cc
(C) 66.7 cc (D) 100 cc

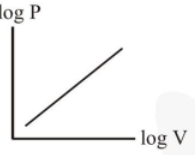
Q9. An open flask containing air is heated from 300K to 500 K. What percentage of air will be escaped to the atmosphere, if pressure is kept constant.
(A) 80 (B) 40
(C) 60 (D) 20

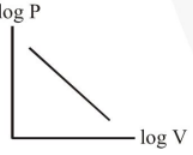
Q10. What temperature would be required to change 6 litres of a gas at 1 atm pressure and 27°C to 10 litres at 0.8 atm pressure ?
(A) 350 K (B) 400 K
(C) 480 K (D) 680 K

Q11. According to Boyle's law, PV = constant. The curve between log P vs log V is :-

(A) 

(B) 

(C) 

(D) 

Q12. If a gas is heated at constant pressure. Its density :

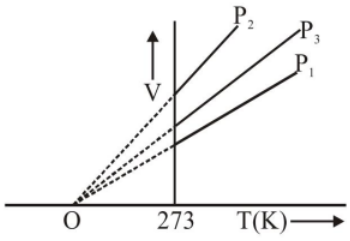
(A) Will increase
(B) Will decrease
(C) Will remain unchanged
(D) May increase or decrease

Q13. For a definite amount of gas pressure and volume are increased to triple of the initial amount. Therefore -

(A) Temperature increased to nine time of its initial value
(B) Temperature increased to thrice of its initial value
(C) Temperature remain unaltered
(D) Temperature reduced to thrice of its initial value

Ideal Gas Equation Part-1

Q14. The volume-temperature graph of a given mass of an ideal gas at constant pressure are shown below :



What is the correct order of pressures

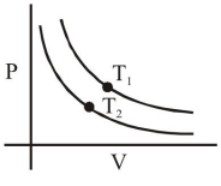
(A) $P_1 > P_3 > P_2$ (B) $P_1 > P_2 > P_3$
(C) $P_2 > P_3 > P_1$ (D) $P_2 > P_1 > P_3$

Q15. The pressure p of a gas is plotted against its absolute temperature T for two different constant volumes, V_1 and V_2. When $V_1 > V_2$, the :-

(A) Curves have the same slope and do not intersect
(B) Curves must intersect at some point other than T = 0
(C) Curve for V_2 has a greater slope than that for V_1
(D) Curve for V_1 has a greater slope than that for V_2

Q16. According to graph, which is true :-

(A) $T_1 > T_2$
(B) $T_1 < T_2$
(C) $T_1 = T_2$
(D) None of these



Q17. At what temperature, 2.8 g of carbon monoxide at 0.821 atm pressure will be 3 litres ?

($R = 0.0821$ L-atm-mol^{-1}-K^{-1})

(A) 350 K (B) 320 K
(C) 300 K (D) 273K

Q18. Densities of a gas at different conditions are in the ratio 1 : 2 and their temperatures are in the ratio 2 : 1, then the ratio of their respective pressure is :-

(A) 1 : 1 (B) 1 : 2
(C) 2 : 1 (D) 4 : 1

Q19. On a ship sailing in pacific ocean where temperature is 23.4°C, a balloon is filled with 2 L air. The volume of the balloon when the ship reaches Indian ocean at a temperature of 26.1°C is -

(A) 2.19 L

(B) 2.17 L

(C) 2.018L

(D) 2.018 ML

Q20. Time taken to distribute one Avogadro number of wheat grains, if 10^{10} grains are distributed in each second is -

(A) 1.90956×10^6 year

(B) 1.3046×10^6 year

(C) 1.90956×10^5 year

(D) 1.3046×10^5 year

Q21. When an open container of volume V is heated from normal temperature T_1 K to T_2 K, the volume of expelled air at temperature T_2 K is $\Delta V'$. Therefore value $\Delta V'/V$ is -

(A) $1 - \frac{T_2}{T_1}$ (B) $1 - \frac{T_1}{T_2}$

(C) $1 + \frac{T_1}{T_2}$ (D) $\frac{T_2}{T_1} - 1$

STP, NTP and Partial Pressure of a Gas Part-2

Q22. A 821 mL $N_2(g)$ was collected over liquid water at 300 K and 1 atm. If vapour pressure of H_2O is 30 torr then moles of $N_2(g)$ in moist gas mixture is :-

(A) 0.39 (B) 0.032

(C) 0.96 (D) 0.0013

Q23. 60% of H_2 and 40% of O_2 by volume are present in a mixture at a pressure of 100 mm. Pressure exerted by H_2 is -

(A) 50 mm (B) 400 mm

(C) 60 mm (D) 1000 mm

Q24. At 27° C a closed vessel contains equal weights of He, CH_4 and SO_2 exerts a pressure of 210 mm. and partial pressures of He, CH_4 and SO_2 are P_1, P_2 and P_3. Which one of the following is correct?

(A) $P_3 > P_2 > P_1$ (B) $P_1 > P_2 > P_3$

(C) $P_1 > P_3 > P_2$ (D) $P_2 > P_3 > P_1$

Q25. A certain mass of a gas occupies a volume of 2 L at STP. To what temperature the gas must be heated to double its volume, keeping the pressure constant?

(A) 100 K (B) 273 K

(C) 273° C (D) 546° C

Problems on Partial Pressure

Q26. The partial pressure of hydrogen in a flask containing 2g of H_2 and 32 g of SO_2 is :-

(A) $\frac{1}{16}$ of total pressure

(B) $\frac{1}{2}$ of total pressure

(C) $\frac{2}{3}$ of total pressure

(D) $\frac{1}{8}$ of total pressure

Q27. 2g H_2 is present with 32 g SO_2 in a 1L containter. If the total pressure of mixture is 2 atm. Then partial pressure would be :

(A) $P_{H_2} = 1$ atm, $P_{SO_2} = 1$ atm

(B) $P_{H_2} = \frac{1}{3}$ atm, $P_{SO_2} = \frac{2}{3}$ atm

(C) $P_{H_2} = \frac{2}{3}$ atm, $P_{SO_2} = \frac{1}{3}$ atm

(D) None

Q28. 56g of nitrogen and 96 g of oxygen are mixed isothermaly and at a total pressure of 10 atm. The partial pressures of oxygen and nitrogen (in atm) are respectively :-

(A) 4, 6 (B) 5, 5

(C) 2, 8 (D) 6, 4

Q29. Equal masses of methane and oxygen are mixed in an empty container at 25°C. The fraction of the total pressure exerted by oxygen is -

(A) 2/3 (B) 1/3

(C) 1/13 (D) 1/2

Graham's law of diffusion

Q30. Calculate relative rate of effusion of O_2 to CH_4 through a container containing O_2 and CH_4 in 3 : 2 mass ratio.

(A) $\frac{3\sqrt{2}}{4}$ (B) $\frac{3}{4\sqrt{2}}$

(C) $\frac{3}{2\sqrt{2}}$ (D) None of these

Q31. Rate of diffusion of A is 5 times that of B, what will be the density ratio of A & B ?

(A) 1 : 25 (B) 1 : 5

(C) 25 : 1 (D) 4 : 1

Q32. 5 moles of O_2 gas diffused through a porous partition in 30 seconds. How many moles of H_2 gas can diffuse through the partition in 60 seconds under the same condition of pressure and temperature?

(A) 2.5 (B) 10

(C) 40 (D) 20

Q33. A rubber balloon permeable to all isotopic forms of hydrogen is filled with heavy hydrogen and placed in tank of pure hydrogen. After some times, the balloon will:

(A) Shrink in size

(B) Expand

(C) Remain as such

(D) Shrink to half of the size

Kinetics of Theory

Q34. Density ratio of O_2 and H_2 molecules is 16 : 1. The ratio of their r.m.s. velocities will be :-

(A) 1 : 4 (B) 1 : 16

(C) 4 : 1 (D) 16 : 1

Q35. At what temperature, r.m.s velocity of O_2 be the same as that of SO_2 at 303 K?

(A) 606 K (B) 303 K

(C) 403 K (D) 273 K

Q36. Which of the following molecule has the largest root mean square velocity at 25° C?

(A) SO_2 (B) CO_2

(C) NH_3 (D) H_2S

Q37. The average speed at temperature T°C of CH_4 (g) is $\sqrt{\frac{28}{88}} \times 10^3$ ms^{-1}. What is the value of T ?

(A) 240.55°C (B) –32.45°C

(C) 3000°C (D) –24.055°C

Q38. Consider a mixture of SO_2 and O_2 kept at room temperature. Compared to the oxygen molecule, the SO_2 molecule will hit the wall with :-

(A) Smaller average speed

(B) Greater average speed

(C) Greater kinetic energy

(D) Greater mass

Q39. The temperature of gas is raised from 27°C to 927°C. The root mean square speed is :-

(A) $\sqrt{\frac{927}{27}}$ times the earlier value

(B) Same

(C) Halfned

(D) Doubled

Q40. According to the kinetic theory of gases, in an ideal gas, between two successive collisions a gas molecule travels -

(A) In a straight line path

(B) with an accelerated velocity

(C) In a circular path

(D) In a wavy path

Q41. The internal energy of one mole of a gas is :-

(A) $\frac{3}{2}$ kT (B) $\frac{1}{2}$kT

(C) $\frac{3}{2}$ RT (D) $\frac{1}{2}$RT

Q42. Kinetic energy of one mole of an ideal gas at 300 K is :-

(A) 3.48 kJ (B) 3.74 kJ

(C) 34.8 kJ (D) 348 kJ

Q43. If r.m.s. velocity of the molecules of a gas is 184.5 m-s^{-1}, then its average velocity is :-

(A) 170 m-s^{-1} (B) 200 m-s^{-1}

(C) 340 m-s^{-1} (D) 400 m-s^{-1}

Q44. To an evacuated vessel with movable piston under external pressure of 1 atm., 0.1 mol of He and 1.0 mol of an unknown compound (vapour pressure 0.68 atm, at 0°C) are introduced. Considering the ideal gas behaviour, the total volume (in litre) of the gases at 0°C is close to -

(A) 7 (B) 9

(C) 8 (D) 10

Collision Theory

Q45. N_2 molecule is spherical of radius 100 pm. What is the volume of molecules of one mole gas? Also find the value of vander waal's constant b?

[Use : $\frac{4}{3}\pi = 4.2$, $N_A = 6 \times 10^{23}$]

(A) 2.52×10^{-3} l mol^{-1}, 10.08×10^{-3} dm^3 mol^{-1}

(B) 2×10^{-3} l mol^{-1}, 10×10^{-3} dm^3 mol^{-1}

(C) 2.62×10^{-3} l mol^{-1}, 9.08×10^{-3} dm^3 mol^{-1}

(D) 2.52×10^{-2} l mol^{-1}, 10.08×10^{-4} dm^3 mol^{-1}

Real gas, Van der waals equation

Q46. The gases deviate from the ideal gas behaviour, because their molecules have :-

(A) negligible volume

(B) forces of attraction

(C) polyatomic structure

(D) constant volume

Q47. Consider the equation $Z = \frac{PV}{nRT}$ which of the following statement is correct ?

(A) When Z > 1 real gases are easier to compress than the ideal gas

(B) When Z = 1 real gases get compressed easily

(C) When Z > 1 real gases are difficult to compress

(D) None

Q48. Vander Walls constant b of helium is 24 mL mol^{-1}. Find molecular diameter of helium.

(A) 1.335×10^{-10} cm

(B) 1.335×10^{-8} cm

(C) 2.67×10^{-8} cm

(D) 4.34×10^{-8} cm

Q49. Vander wall's equation for $\frac{1}{2}$ mole of gas :-

(A) $\left(P + \frac{a}{V^2}\right)(V - b) = RT$

(B) $\left(P + \frac{a}{4V^2}\right)\left(V - \frac{b}{2}\right) = \frac{RT}{2}$

(C) $\left(P+\frac{a}{4V^2}\right)\left(\frac{V-b}{2}\right)=RT$

(D) $\left(P+\frac{a}{4V^2}\right)\left(\frac{V-b}{2}\right)=2RT$

Q50. The value of vander Waals' constant 'a' for gases O_2, N_2, NH_3 and CH_4 are 1.360, 1.390, 4.170 and 2.253 litre^2 atm mol^{-2} respectively. The gas which can be most easily liquefied is :-

(A) O_2 (B) N_2

(C) NH_3 (D) CH_4

Q51. A real gas may behave as ideal gas at :-

(A) 2000 K, 200 atm

(B) 2 atm, 2000 K

(C) 100 K, 2 atm

(D) 200 atm, 100 K

Q52. Vander Waal's equation of state is oberyed by real gases. For n moles of a real gas, the expression will be :-

(A) $\left(\frac{P}{n}+\frac{na}{V^2}\right)\left(\frac{V}{n-b}\right)=RT$

(B) $\left(P+\frac{a}{V^2}\right)(V-b)=nRT$

(C) $\left(P+\frac{na}{V^2}\right)(nV-b)=nRT$

(D) $\left(P+\frac{n^2a}{V^2}\right)(V-nb)=nRT$

Q53. At high temperature and low pressure, the Vander Waal's equation is reduced to :-

(A) $\left(p+\frac{a}{V_m^2}\right)(V_m)=RT$

(B) $pV_m=RT$

(C) $P(V_m-b)=RT$

(D) $\left(p+\frac{a}{V_m^2}\right)(V_m-b)=RT$

Q54. A real gas obeying vander wall's equation will resemble ideal gas if constants :-

(A) a and b are small

(B) a is large and b is small

(C) a is small and b is large

(D) a and b are large

Q55. The gases show significant deviations from ideal gas laws at :-

(A) Low pressure (B) High pressure

(C) Low temperature (D) Both 'B' and 'C'

Real gas, Compressibility Factor

Q56. 0.5 moles of a gas occuping V volume at P pressure and T kelvin temperature. The compressibility factor Z is equal to :-

(A) $\frac{PV}{RT}$ (B) $\frac{2PV}{RT}$ (C) $\frac{PV}{2RT}$ (D) $\frac{RT}{PV}$

Q57. The compressibility factor for a real gas showing positive deviation is :-

(A) $Z<0$ (B) $Z>1$

(C) $Z>0$ (D) $Z<1$

Q58. The compressibility factor for a real gas at high pressure is :-

(A) 1 (B) $1+\frac{pb}{RT}$

(C) $1-\frac{pb}{RT}$ (D) $1+\frac{RT}{pb}$

Q59. At low pressure, the Vander Waal's equation is reduced to :-

(A) $Z=\frac{pV_m}{RT}=1-\frac{ap}{RT}$

(B) $Z=\frac{pV_m}{RT}=1-\frac{a}{V.RT}$

(C) $pV_m=RT$

(D) $Z=\frac{pV_m}{RT}=1-\frac{a}{RT}$

Liquifaction of gases

Q60. Under critical conditions, the compressibility factor for a gas is :-

(A) $\frac{3}{8}$ (B) $\frac{8}{3}$ (C) 1 (D) $\frac{1}{4}$

Q61. The Vander Waal's parameters for gases W, X, Y and Z are :-

Gas	a (atm L^2 mol^{-2})	b (L mol^{-1})
W	4.0	0.027
X	8.0	0.030
Y	6.0	0.032
Z	12.0	0.027

Which one of these gases has the highest critical temperature ?

(A) W (B) X

(C) Y (D) Z

Liquid state

Q62. Following are the critical temperature of the some gases :

Gases	H_2	He	O_2
T_C(K)	33.2	5.3	154.3

From the above data what would be order of liquification of these gases.

Start writing the order from the gas liquifying first

(A) H_2, He, O_2 (B) He, O_2, H_2

(C) O_2, He, H_2 (D) O_2, H_2, He

ACC

Q63. Molecular weight of a gas that diffuses twice as rapidly as the gas with molecular weight 64 is :-

(A) 16 (B) 8 (C) 64 (D) 6.4

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which are correct statements about pressure?

(A) It's S.I unit is Pascal

(B) 1 bar = 0.986 atm

(C) 1 torr = 720 mm of Hg

(D) 1 atm = 101.325 KN/m^2

Q2. If the pressure of the gas contained in a closed vessel is increased by 20% when heated by 273°C then it's initial temperature must have been :

(A) 2184 °C (B) 1092 °C

(C) 1365 K (D) 2457 K

Q3. Absolute zero temperatue is -

(A) Zero Kelvin (B) – 273° C

(C) 273 K (D) – 273 K

Q4. Which of the following will represent graph of y = mx + C for ideal gas ?

(A) Pressure versus Volume at constant temperature.

(B) Pressure versus $\frac{1}{\text{Volume}}$ at constant temeperature (0°C)

(C) Volume versus temperature (°C) at constant pressure

(D) Pressure versus temperature (°C) at constant volume

Q5. Which of the following is/are incorrect for the Barometer?

(A) Pressure measured is dependent on height of mercury raised.

(B) It is a long cylindrical tube

(C) Pressure measured depend on density of liquid filled

(D) Pressure measured is dependent on cross sectional area of tube

Q6. Which is true about NTP?

(A) Temperature will be 273 K

(B) Pressure will be 1 Bar

(C) For 1 mole gas it occupies 22.4 Litre

(D) For 1 mole of gas it occupies 22.7 Litre

Q7. Pressure of a gas in closed container can be increased by

(A) Reducing volume of container

(B) By increasing temperature of container

(C) By cooling the gas

(D) By adding more gas into it

Q8. Which statement is correct for the following?

(A) $U_{rms} = \sqrt{\frac{3RT}{M}}$ (B) $U_{avg} = \sqrt{\frac{8RT}{\pi M}}$

(C) $U_{avg} = \sqrt{\frac{3RT}{M}}$ (D) $U_{rms} = \sqrt{\frac{8RT}{\pi M}}$

Q9. Which does not follow graham's law, for rate of diffusion $\frac{r_A}{r_B}$?

(A) $\frac{P_A}{P_B}\left(\frac{M_A}{M_B}\right)^{1/2}$ (B) $\left(\frac{M_A}{M_B}\right)\left(\frac{P_A}{P_B}\right)^{1/2}$

(C) $\left(\frac{P_A}{P_B}\right)\left(\frac{M_B}{M_A}\right)^{1/2}$ (D) $\left(\frac{M_A}{M_B}\right)\left(\frac{P_B}{P_A}\right)^{1/2}$

Q10. Which of the following pairs have different rate of diffusion?

(A) NO, CO (B) NO, C_2H_6

(C) N_2, C_2H_4 (D) NH_3, PH_3

Q11. Which of the following is **incorrect** order for effusion of gases H_2, O_2, NH_3 and CO_2 at constant P & T is-

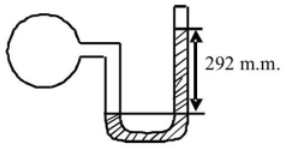
(A) $H_2 > CO_2 > NH_3 > O_2$

(B) $H_2 > NH_3 > O_2 > CO_2$

(C) $H_2 > O_2 > NH_3 > CO_2$

(D) $CO_2 > NH_3 > O_2 > H_2$

Q12. An open ended Hg manometer is used to measure the pressure exerted by a trapped gas as shown in the figure. Atmospheric pressure is 749 m.m. of Hg. What is the pressure of the trapped gas-



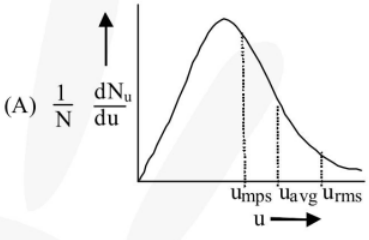
(A) 749 m.m. of Hg

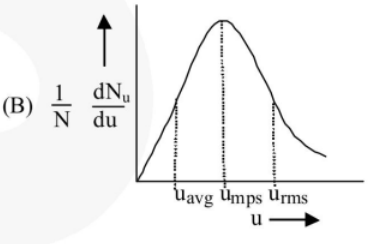
(B) 1041 m.m. of Hg

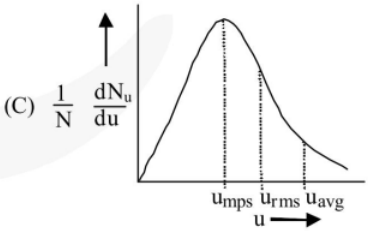
(C) 1.37 atm

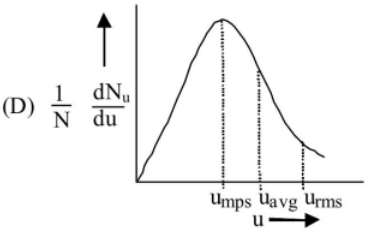
(D) 749 atm

Q13. Which of the following plot is not correct ?

(A) 

(B) 

(C) 

(D) 

Q14. According to kinetic theory of gases

(A) collisions are always elastic

(B) heavier molecules transfer more momentum to the wall of the container

(C) only a small number of molecules have very high velocity.

(D) between collisions, the molecules move in straight lines with constant velocities.

Q15. The kinetic energy of one mole of any gas is independent from.

(A) Pressure of gas

(B) Volume of gas

(C) Nature of gas

(D) Absolute temperature of gas

Q16. The RMS velocity of 1 mole of gas at NTP can be calculated as -

(A) $\sqrt{\frac{3P}{d}}$ (B) $\sqrt{\frac{3PV}{M}}$

(C) $\sqrt{\frac{PV}{3M}}$ (D) $\sqrt{\frac{3RT}{M_w}}$

Q17. Select the **correct** option(s) for an ideal gas

(A) Most probable speed increases with increase in temperature

(B) Fraction of particles moving with most probable speed increases with increase in temperature

(C) Fraction of particles moving with most probable speed are more for Cl_2 than H_2 under similar condition of T, P & V.

(D) Most probable speed is more for Cl_2 than H_2 at same temperature

Q18. Which of the following statements are true for a gas -

(A) The mean free path is inversely proportional to the pressure of the gas at constant temperature.

(B) The mean free path is proportional to temperature at constant pressure.

(C) The collision frequency among the molecules of gas is proportional to temperature.

(D) The velocity possessed by the largest fraction of molecules at a given temperature is known as root mean square velocity.

Q19. The density of CO at 273 K and 1 atm is 1.2504 kgm^{-3}, then.

(A) Root means sqaure speed is = 493 m/s

(B) Most probable speed is = 403 m/s

(C) Average speed = 454.4 m/s

(D) All will have the same value

Q20. According to Maxwell - Boltzman distribution of speeds among gas molecules, what is false?

(A) The maxima in plot of $\frac{\Delta N}{N}$ vs speeds pertains to average speed

(B) The plot of $\frac{\Delta N}{N}$ vs speeds is straight line with slope > 0

(C) The value of r.m.s increases with rise in temperature.

(D) Area under the curve gives the total number of molecules.

Q21. In the real gas equation -

(A) Higher the value of 'a' more easily a gas can be liquified

(B) More the value of 'b' more will be molecular size

(C) Unit of a = $litre^2$ atm mol^{-2} and b = litre mol^{-1}

(D) a and b are unit less

Q22. Which of the following is true statement about vander waal contant 'b' from the given statement?

(A) It's called excluded volume

(B) It account for interparticle forces

(C) Its unit is mol dm^{-3}

(D) It's value depend on molecular size

Q23. In the graph of compressibility factor versus pressure which is/are correct?
(A) Value of Z for H_2 & He is always greater than 1
(B) For other real gas value of Z first decrease and then increase with increase in pressure.
(C) Z = 1 for ideal gas
(D) Z is independent of pressure

Q24. Which of the following expressions of compression factor Z (= pV_m / RT) of a real gas is false at low pressure -
(A) $Z = 1 - a / V_m RT$ (B) $Z = 1 + a / V_m RT$
(C) $Z = 1 + pb / RT$ (D) $Z = 1 - pb / RT$

Q25. The condition(s) in which a gas can be easily liquified is/are -
(A) compressibility factor is more than 1
(B) the value of van der Waal's constant 'a' is low.
(C) the value of van der Waal's constant 'a' is high.
(D) below critical temperature at high pressure

Q26. Identify the incorrect statement :-
(A) Solids are easily compressible
(B) Liquids will have indefinite volume but definite shape
(C) Gases have highest degree of motion
(D) Substance having Boiling point above room temperature will exist as gas in a room temperature.

Q27. A gas can be liquified by -
(A) Cooling
(B) Compressing
(C) Heating
(D) Decreasing pressure

Q28. Which of the following is true for Hg to be used in barometer?
(A) It is a high density liquid
(B) It is viscous liquid
(C) It shows capillary action
(D) It has low boiling point than water

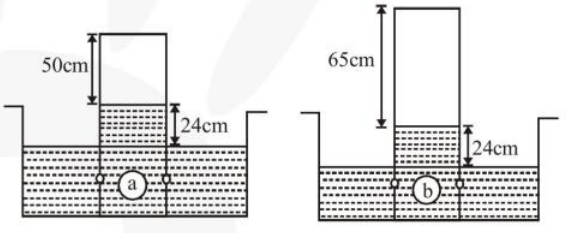
Q29. Select the incorrect statement(s) :
(A) At Boyle's temperature a real gas behaves like an ideal gas irrespective of pressure.
(B) At critical condition, a real gas behaves like an ideal gas.
(C) On increasing the temperature four times, collision frequency (Z_{11}) becomes double at constant volume.
(D) At high pressure Vander Waal's constant **'b'** dominates over **'a'**.

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph For Q. 30 to 32

Same amount of a gas is stored over liquid (a) and liquid (b) as shown in diagram.



	Density	V. pressure of liquid
Liquid (a)	3.4 gm/ml	50mm of Hg
Liquid (b)	–	110mm of Hg

[Given : Atmospheric pressure = 760mm, Density of mercury = 13.6gm/ml]

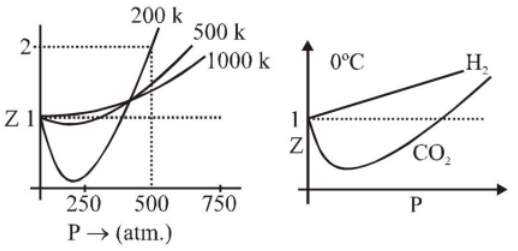
Q30. The pressure (in cm of Hg) of the gas over liquid (a) is –
(A) 65 (B) 70 (C) 76 (D) 59

Q31. The pressure (in cm of Hg) of the gas over liquid (b) is –
(A) 56 (B) 65
(C) 50 (D) None of these

Q32. Density of liquid (b) is –
(A) 10gm/ml (B) 5 gm/ml
(C) 2.5 gm/ml (D) None of these

Paragraph for Q. 33 to 35

Sketch shows the plot of Z v/s P for of a hypothetical gas for one mole at three distinct temperature.



Boyle's temperature is the temperature at which a gas shows ideal behaviour over a pressure range in the low pressure region. Boyle's temperature (T_b) = $\frac{a}{Rb}$. If a plot is obtained at temperatures well below Boyle's temperature then the curve will show negative deviation, in low pressure region and positive deviation in the high pressure region. Near critical temperature the curve is more likely as CO_2 and the temperature well above critical temperature curve is more like H_2 at 0°C as shown above. At high pressure suppose all the constant temperature curve varies linearly with pressure according to the following equation:

$$Z = 1 + \frac{Pb}{RT} \qquad (R = 2 \text{ cal mol}^{-1} \text{ K}^{-1})$$

Q33. Which of the following is correct :

(A) $\frac{a}{b} < 0.4$ k cal mol^{-1}

(B) 0.4 k cal mol^{-1} $< \frac{a}{b} <$ 2 k cal mol^{-1}

(C) $\frac{a}{b} < 0.2$ k cal mol^{-1}

(D) $\frac{a}{b} = 1$ K cal mol^{-1}

Q34. For 500 K plot value of Z changes from 2 to 2.2 if pressure is varied from 1000 atm to 1200 atm (high pressure) then the value of $\frac{b}{RT}$ will be :

(A) 10^{-3} atm^{-1} (B) 2×10^{-3} atm^{-1}

(C) 5×10^{-4} atm^{-1} (D) 10^{-4} atm^{-1}

Q35. As shown in the figure at 200 K and 500 atm value of compressibility factor is 2 (approx). Then volume of the gas at this point will be :

(A) 0.01 L (B) 0.09 L

(C) 0.065 L (D) 0.657 L

Paragraph For Q. 36 to 39

Real gas obey the equation of state PV = nRT under condition of low pressure and high temperature. In general, the most easily liquefiable gases, like CO_2, SO_2 and ammonia show much longer deviations from ideal behaviour than gases like H_2, N_2, Or, O_2. This can be seen from the compressibility factor (Z) – pressure diagram for various gases. the deviation from ideal behaviour is attributed to two factor (i) The volume occupied by gas molecules is not negligible in comparison to the total volume of gas (ii). The intermolecular forces between gas molecules are significant. To account for these deviation, the vander Waals equation has been proposed.

Q36. The vander Waals constant "a" for SO_2, ethylene, HCl and CO_2 are 6.71, 4.47, 3.67 and 3.59 L^2 atm mol^{-2} respectively. The order of liquefaction of the gases increase in the order -

(A) $CO_2 < HCl < C_2H_4 < SO_2$

(B) $SO_2 < C_2H_4 < HCl < CO_2$

(C) $C_2H_2 < SO_2 < CO_2 < HCl$

(D) $HCl < SO_2 < C_2H_4 < CO_2$

Q37. The critical compressibility factor Z_C for O_2 is 0.308 and its critical pressure and critical volume are 50.1 atm and 7.8×10^{-2} litre mol^{-1} respectively. The Boyle temperature of O_2 is - (in deg K)

(A) 348.6 (B) 215.4

(C) 154.5 (D) 521.4

Q38. At extremely low pressures, the van der Waals equation for one mole of a gas may be expressed as -

(A) $PV = RT - \frac{a}{v}$

(B) $P(V - b) = RT$

(C) $(P + a)(V - b) = RT$

(D) $PV = RT$

Q39. A gas obeys the equation of state $P(V_m - b) = RT$.

The slope for an isochore will be -

(A) Negative

(B) Zero

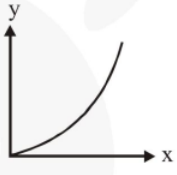
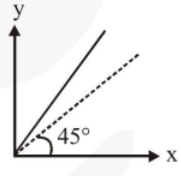
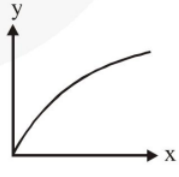
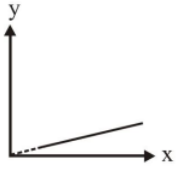
(C) $\frac{R}{(V_m - b)}$

(D) R/P

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q40. Given that in all cases the parameters before v/s to be on y-axis and the parameter after v/s to be on x-axis. All the parameter are taken and plotted in SI system.

Column I	**Column II**
(A) PT v/s T^2 [n & V constant]	(P) 
(B) PV v/s T $\left[n = \frac{1}{10}\text{mol}\right]$	(Q) 
(C) $\frac{P}{V}$ v/s $\frac{1}{V^2}$ [n = 1 mol and T = 298 K]	(R) 
(D) V v/s $\sqrt{T}$ [n & P constant]	(S) 

Q41.

	Column I (Gases X and Y taken of diffusion)	**Column II (Ratio of times taken)**
A.	X = 100ml of H_2 at 1 bar, 25°C Y = 200ml of O_2 at 1 bar, 25°C	P. 1 : 1.225
B.	X = 100ml of O_2 at 1 bar, 25°C Y = 200ml of O_3 at 2 bar, 25°C	Q. 1 : 0.7
C.	X = 100ml of SO_2 at 1 bar, 25°C Y = 100ml of O_2 at 1 bar, 25°C	R. 1 : 1.36
D.	X = HCl gas to travel 100 cm length in a tube Y = NH_3 gas to travel 200 cm length using the same tube (P,V,T = same in both cases)	1 : 8

Q42. Four different gases and their vanderwaal's constant are as given-

Gas	a in atm L^2mol^{-2}	b in L/mol
A_2	1.5	0.05
B_2	2.0	0.08
C_2	4.2	0.025
D_2	5.0	0.1

Column-I	**Column-II**
(A) Gas having highest inter-molecular force of attraction.	(P) A_2
(B) Gas having smallest size of molecules.	(Q) B_2
(C) Gas that can be liquefied at 90 K.	(R) C_2
(D) Gas resembles with ideal gas at the boiling point of water	(S) D_2

Q43. Match gases under specific conditions listed in Column I with their properties/laws in Column II.

Column I	**Column II**
(A) Hydrogen gas (P = 200 atm, T = 273 K)	(P) Compressibility factor ≠ 1
(B) Hydrogen gas (P ~ 0, T = 273 K)	(Q) Attractive forces are dominant
(C) CO_2 (P = 1 atm, T = 273 K)	(R) PV = nRT
(D) Real gas with very large molar volume	(S) P (V – nb) = nRT

Q44.

Column I	**Column II**
(A) Excluded molar volume	(P) Z = 1
(B) Higher the compressibility factor	(Q) b
(C) At Boyle's temperature	(R) All gases behave ideally
(D) At extremely low pressure	(S) Difficult is the liquification of gas

Q45. 6 litre H_2O is placed in a closed room of volume 827 L at the temperature of 300K. If vapour pressure of liquid water is 22.8mm of Hg at 300 K and its density is 1 g/cm^3.

Given R = 0.0821 atm L mol^{-1} K^{-1}, Assuming volume of liquid water to be constant]

Column I	Column II
(A) Mass of H_2O in gaseous form (in gm)	(p) 6
(B) Mass of H_2O in gaseous state (in moles)	(q) 18
(C) Approximate amount of water left in liquid state (in kg)	(r) 3
(D) Total number of moles of all atoms in vapour form	(s) 1

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q46. **Statement 1 :** Time taken for the effusion of same moles of H_2(g) is greater than that of He(g), if $(r_{He} / r_{H_2}) = 2$.

[where r_{H_2} and r_{He} are rate of effusion of H_2(g) and He(g) respectively]

Statement 2 : Molecular mass of He(g) is greater than that of molecular mass of H_2(g).

Q47. **Statement 1 :** 1/4th of gas is expelled if air present in an open vessel is heated from 27°C to 127°C.

Statement 2 : Rate of diffusion of a gas is inversely proportional to the square root of its molecular mass.

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q48. If 'X' represent no. of substance in solid state at room temperature. 'Y' represent no. of substance in liquid at room temperature. 'Z' in gaseous state at room temperature. Find value of X + Y – Z from following - urea, Ammonia Kerosene, Water, Acetic acid, Bromine, Mercury.

Q49. The value of universal gas constant in calorie k^{-1} mol^{-1}?

Q50. One mole of an ideal gas is subjected to a process in which P (in atm) = $\frac{1}{8.21}$V (in Litre). If the process is operating from a initial pressure 1 atm to final pressure 10 atm (no higher pressure achieved during the process) then what would be the maximum temperature obtained & at what instant will it occur in the process.

Q51. A gas is present in a cylinder fitted with movable piston. Above and below of the piston there is equal number of moles of gas. The volume above is two times the volume below at a temperature of 300K. At what temperature will the volume above be four times the volume below.

Q52. A gaseous mixture contains 1 part of H_2 in every 65 parts by mass of a mixture of H_2 and O_2 gases. How many diffusion steps is needed such that the final gaseous mixture coming out should have 1 part of H_2 in every 5 parts by mass of the mixture?

Q53. By how many times the absolute temperature of a gas increased when U_{rms} of a gas in a container of fixed volume is increased from 5×10^4 cm/sec to 10×10^4 cm/sec.

Q54. The average velocity of gas molecule is 400 m/sec. It's rms at same temperature will be-

Q55. If the mean free path of gaseous molecules is 60 cm at a pressure of 1×10^{-4} mm Hg, what will be its mean free-path when the pressure is increased by 100 times ?

Q56. For a vessel at 1832 K containing 10 moles of steam at 50 atm. volume would be :
(Given : a = 5.46 atm. $L^2 mol^{-2}$, b = 0.031 L mol^{-1}) -

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. A jar contains a gas and a few drops of water. The pressure in the jar is 830 mm of Hg. The temperature of the jar is reduced by 1%. The vapour pressure of water at two temperatures are 30 and 25 mm of Hg. Calculate the new pressure in Jar.
(A) 792 mm of Hg
(B) 817 mm of Hg
(C) 800 mm of Hg
(D) 840 mm of Hg

Q2. If the pressure of a given mass of gas is reduced to half and temperature is doubled simultaneously, the volume will be-
(A) Same as before
(B) Twice as before
(C) Four time as before
(D) One fourth as before

Q3. The pressure-volume plot for an ideal gas at a given temperature has the form of a
(A) Straight line
(B) Exponential curve
(C) Rectangular hyperbola
(D) U-shaped curve

Q4. At STP, a container has 1 mole of He, 2 mole Ne, 3 mole O_2 and 4 mole N_2. Without changing total pressure if 2 mole of O_2 is removed, the partial pressure of O_2 will be decreased by :-
(A) 26% (B) 40%
(C) 58.33% (D) 66.66%

Q5. Which of the following mixture when burnt to produce $CO_2(g)$ and $H_2O(\ell)$, will cause a 40% reduction in volume
(A) 10ml $C_3H_6O_2$ + 50ml O_2
(B) 40ml $C_2H_2O_4$ + 60ml O_2
(C) 10ml C_3H_8 + 110ml O_2
(D) 20ml C_2H_6O + 80ml O_2

Q6. Two closed vessels of equal volume containing air at pressure p_1 and temperature T_1 are connected to each other through a narrow tube. If the temperature in one of the vessels is now maintained at T_1 and that in the other at T_2, what will be the pressure in the vessels :-
(A) $\frac{2P_1T_1}{T_1+T_2}$ (B) $\frac{T_1}{2P_1T_2}$
(C) $\frac{2P_1T_2}{T_1+T_2}$ (D) $\frac{2p_1}{T_1+T_2}$

Q7. The rate of effusion of helium gas at a pressure of 1000 torr is 10 torr min^{-1} What will be the rate of effusion of hydrogen gas at a pressure of 2000 torr at the same temperature?
(A) 20 torr min^{-1} (B) 40 torr min^{-1}
(C) $20\sqrt{2}$ torr min^{-1} (D) 10 torr min^{-1}

Q8. A spherical air bubble is rising from the depth of a lake when pressure is P atm & temperature is T K. The percentage increase in its radius when it comes to the surface of lake will be (Assume temperature & pressure at the surface to be respectively 2 T K & P/4)

(A) 100 % (B) 50 %

(C) 40 % (D) 200 %

Q9. An open end manometer consists of 0.75 moles of $X_{(g)}$ taken in a container of volume 24.63 lit. at 300 K. The level of mercury in the open tube is found to be 14 cm lower. The height difference when manometer is heated to 450 K will be :

(A) 14.5 cm (B) 21 cm

(C) 13 cm (D) 17 cm

Q10. Equal volumes of SO_2 and He at a temperature T and pressure P are allowed to effuse through a hole. The rate of effusion of helium is -

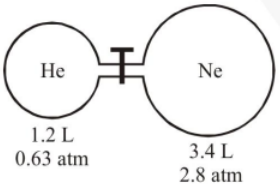
(A) Equal to the rate of effusion of SO_2

(B) Four times the rate of effusion of SO_2

(C) Half of the rate of effusion of SO_2

(D) Twice the rate of effusion of SO_2

Q11. Consider the following apparatus. Calculate the partial pressure of helium after the opening valve.The temperature remains constant at 16°C.



(A) 0.164 atm (B) 1.64 atm

(C) 0.328 atm (D) 1 atm

Q12. As the temperature is raised from 20°C to 40°C, the average kinetic energy of neon atom changes by a factor of which of the following?

(A) 2 (B) $\frac{1}{2}$

(C) $\frac{313}{293}$ (D) $\sqrt{\frac{313}{293}}$

Q13. A gas mixture consists of 2 moles of oxygen and 4 moles of a argon at temperature T. Neglecting all vibrational modes, the total internal energy of the system is –

(A) 4RT (B) 5RT

(C) 15RT (D) 11RT

Q14. Indicate the incorrect statement for a 1-L sample of $N_2(g)$ and CO_2 (g) at 298 K and 1 atm pressure :-

(A) The average translational KE per molecule is the same in N_2 and CO_2

(B) The rms speed remains same for both N_2 and CO_2

(C) The density of N_2 is less than that of CO_2

(D) The total translational KE of both N_2 and CO_2 is the same

Q15. The rms speed of N_2 molecules in a gas is u. If the temperature is doubled and the nitrogen molecules dissociate into nitrogen atoms, the rms speed becomes :

(A) u/2 (B) 2u

(C) 4u (D) 14u

Q16. Kinetic energy of one mole of an ideal gas at 300 K is :-

(A) 3.48 kJ (B) 3.74 kJ

(C) 34.8 kJ (D) 348 kJ

Q17. The rms speed of a gas molecules at temperature 27 K and pressure 1.5 bar is 1×10^4 cm/sec. If both temperature and pressure are raised three time, the rms speed of the gas will be-

(A) 9×10^4 cm/sec (B) 3×10^4 cm/sec

(C) $\sqrt{3} \times 10^4$ cm/sec (D) 1×10^4 cm/sec

Q18. A helium atom is two times heavier than a hydrogen molecule at 298 K, the average kinetic energy of helium is-

(A) Two times that of hydrogen molecules

(B) Same as that of hydrogen molecules

(C) Four times that of hydrogen molecules

(D) Half that of hydrogen molecules

Q19. The root mean square velocity of an ideal gas at constant pressure varies with density as –

(A) d^2 (B) d

(C) $d^{1/2}$ (D) $1/d^{1/2}$

Q20. A container containing 2 moles of H_2 at temperature T has total Z bimolecular collision per second. The total number of bimolecular collisions if same container is filled with 4 moles of H_2 at 2T temperature is –

(A) 2Z (B) 4Z

(C) Z/2 (D) $\sqrt{2}$ Z

Q21. Positive deviation from ideal behaviour takes place because of –

(A) molecular interaction between atoms and $\frac{PV}{nRT} > 1$

(B) molecular interaction between atoms and $\frac{PV}{nRT} < 1$

(C) finite size of atoms and $\frac{PV}{nRT} > 1$

(D) finite size of atoms and $\frac{PV}{nRT} < 1$

Q22. Vander wall's equation for :-

(A) Any pressure

(B) low pressure

(C) force of attraction is negligible

(D) Volume of molecules is negligible

(P) $PV = RT + Pb$

(Q) $PV = RT - \frac{a}{v}$

(R) $PV = RT + \frac{a}{v}$

(S) $\left(p + \frac{a}{v^2}\right)(v - b) = RT$

(A) A → S, B → Q, C → P, D → R

(B) A → Q, B → S, C → P, D → R

(C) A → S, B → Q, C → R, D → P

(D) None

Q23. The critical temperature and critical pressure of a gas obeying van der Waals' equation are 30°C and 73 atm respectively. Its van der Waals' constant, b (in litres mol^{-1}) is, therefore :

(A) 0.500 (B) 0.060

(C) 0.265 (D) 0.042

Q24. The influential parameter (according to Vander Waal's equation) of critical point is-

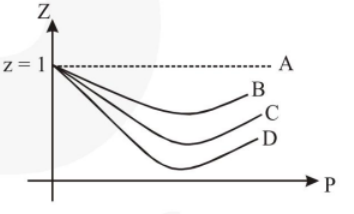
(A) b only

(B) a only

(C) a & b both

(D) Neither a, nor b

Q25. Z vs P graph of four gases is given below at same temperature



Which of the gases will have the highest value of critical temperature.

(A) D (B) C

(C) B (D) A

Q26. However greater the pressure, a gas cannot be liquefied above :-

(A) dry temperature

(B) Inversion temperature

(C) Critical temperature

(D) Room temperature

Q27. The average, RMS and most probable speed of gas molecules at STP increase in the order.

(A) RMS < average speed < most probable speed

(B) Most probable speed < average speed < RMS

(C) Average speed < RMS < most probable speed

(D) RMS < most probable speed < average speed

Q28. A gaseous reaction, $3A \rightarrow 2B$ is carried out in a 0.0821 litre closed container initially containing 1 mole of gas A. After sufficient time a curve of P (atm) as T (K) is plotted and the angle with x-axis was found to be 42.95°. The degree of association of gas A is (Given tan 42.95 = 0.8]

(A) 0.4 (B) 0.6

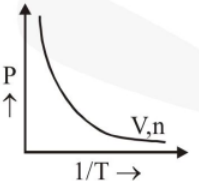
(C) 0.5 (D) 0.8

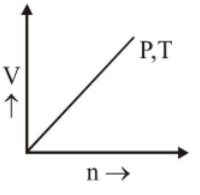
Exercise - 2A — *Brain Booster*

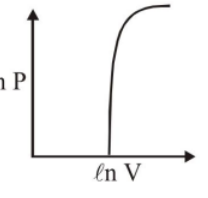
ONE OR MORE THAN ONE CORRECT TYPE

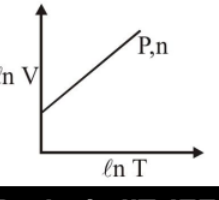
*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following graphs are correct for an ideal gas ?

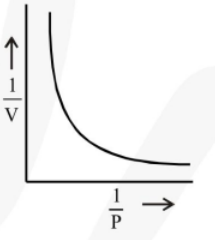
(A)


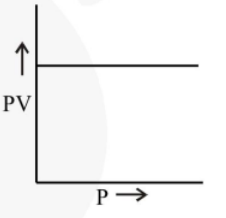
(B)


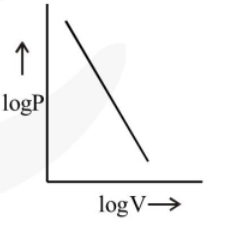
(C)


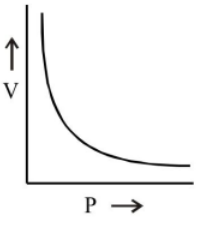
(D)


Q2. Which of the following graph is / are correct for Boyle's law for ideal gas :-

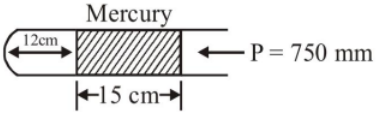
(A)


(B)


(C)


(D)


Q3. Consider the following statement for the given diagram.



(A) Length of air Column is 10 cm if tube is held vertically with open end up.

(B) Length of air Column is 15 cm if tube is held vertically with open end down.

(C) Length of air Column is 10.5 cm if tube is held at 45° with open end up

(D) Length of air Column is 20 cm if tube is held vertically with open end up

Q4. A 1.0 g sample of air consists of approximately 0.76 g of nitrogen and 0.24 g of oxygen. This sample occupies a 1.0 L vessel at 20°C. Then -

(A) The partial pressure of N_2 is 0.65 atm.

(B) The partial pressure of O_2 is 0.36 atm.

(C) The total pressure is 0.83 atm.

(D) The total pressure is 1.05 atm.

Q5. Carbon monoxide gas is purchased in a 425 mL bottle at a pressure of 5 atm. and 23°C. Choose the correct alternatives-

(A) Mass of the gas purchased is 1.45 g

(B) Mass of the gas purchased is 2.45 g

(C) Density of the gas in the bottle is 5.76 g/L

(D) Density of the gas in the bottle is 2.88 g/L.

Q6. A gaseous mixture containing equal moles of He, SO_2 and X (molecular mass = 256) is subjected to series of effusion steps. Select the correct options (s) –

(A) The composition of effused mixture after 4 effusion steps is 4096 : 16 : 1 (by mol)

(B) The composition of effused mixture after 6 effusion steps is 262144 : 64 : 1 (by mol)

(C) Composition of effused mixture cannot be 512 : 8 : 1 after any number of effusing steps

(D) All above options are correct

Q7. If the rms velocity of nitrogen and oxygen molecule are same at two different temperature and same pressure than -

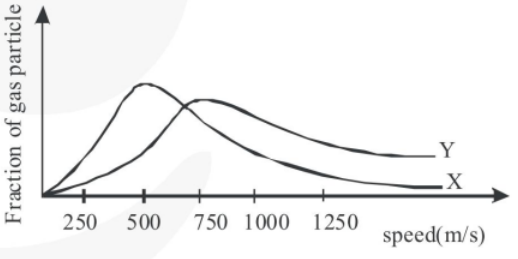
(A) Average speed of molecules is also same

(B) density (gm/It.) of nitrogen and oxygen is also equal.

(C) Number of moles of each gas is also equal.

(D) most probable velocity of molecules is also equal.

Q8. The graph below shows the distribution of molecular speed of two ideal gases X and Y at 200K.on the basis of the below graph identify the correct statements -



(A) If gas X is CO_2 gas Y can be CH_4

(B) Fraction of molecules of X must be greater than Y in a particular range of speed at 200K

(C) Under identical conditions rate of effusion of Y is greater than that of X

(D) The molar kinetic energy of gas X at 200K is equal to the molar kinetic energy of Y at 200K

Q9. A modified form of Vander Waal's equation of state for 1.0 mole of gas is given as :

$\left(P+\frac{\alpha}{TV^2}\right)(V-\beta)=RT$, where α & β are constant.

Select correct statements -

(A) The second virial coefficient B is $\left(\beta-\frac{\alpha}{RT^2}\right)$

(B) The Boyle's temperature T_B is $\sqrt{\frac{\beta}{R\alpha}}$

(C) The second virial coefficient B is $\left(\beta+\frac{\alpha}{RT^2}\right)$

(D) The Boyle's temperature T_B is $\sqrt{\frac{\alpha}{R\beta}}$

Q10. The vander waals constants for HCl are a = 371.843 K Pa dm^2/mol^2 and b = 40.8 $cm^3\ mol^{-1}$. Choose the correct options -

(A) The critical pressure is 8.273 MPa

(B) the critical temperature is 224.8 K

(C) The critical volume is 122.4 cm^3

(D) the critical temperature is 324.8 K

Q11. One mole triatomic vapours of an unknown substance effuses $\frac{4}{3}$ times faster than 1 mole O_2 under same conditions. If the density of unknown vapours at pressure P and temperature T is d, which of the following holds true for the unknown substance -

(A) $d_{N.T.P} = 0.8035\ g/\ell$

(B) Z (atomic number) = 6

(C) Z(compressibility factor) = $\frac{18P}{dRT}$

(D) Vapour density = 9

Q12. Select the correct statements(s)

(A) At Boyle's temperature a real gas behaves like an ideal gas at low pressure

(B) Above critical conditions, a real gas can not liquified

(C) For hydrogen gas 'a' is very large

(D) At high pressure, pressure b domainates over 'a'

Q13. Select the correct statement:

(A) The value of compressibility factor 'Z' for H_2 gas is greater than one at room tempera ture and pressure.

(B) The real gas behaves as an ideal gas at Boyle's temperature, under low pressure region.

(C) For a real gas following van der Waals' equation of state, the expression of critical temperature is $\frac{8a}{27R.b}$.

(D) At low pressure, the compressibility factor 'Z' = $1+\frac{P.b}{RT}$ for a van der Waal's gas.

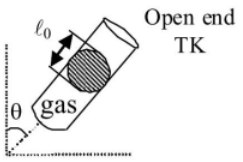
PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph For Q. 14 to 17

A gas column is trapped by the Hg column of length ℓ_0. Atmospheric pressure is P_0. If the tube is held at an angle θ with the vertical then pressure of the gas can be given as -

$$P_{gas} = P_0 + (\ell_0 \cos\theta)\,\rho_{Hg}\,g$$



ρ_{Hg} = 13.6 gm/ml ; g = 10 m/s^2

Gas is ideal in nature and on rotating the glass tube, temperature of the system remain constant.

Q14. If θ = 0° and P_0 = 1 atm, ℓ_0 = 8 cm, then pressure of the gas is-

(A) 840 m.m. of Hg

(B) 1 atm

(C) 680 m.m. of Hg

(D) 1.01 atm

Q15. If θ = 90°, if volume of the gas column is 8.21 L and mass of the gas is 8 gm and temperature of the gas is 50 K, therefore molecular wt. of the gas (monoatomic) will be -

(A) 4 (B) 2

(C) 8 (D) 1

Q16. For θ = 180°, the pressure of the gas will be :-

(A) 760 m.m. of Hg

(B) 680 m.m. of Hg

(C) 84 cm.of Hg

(D) 86 cm. of Hg

Q17. For θ = 45°, the pressure of the gas will be-

(A) 760 m.m. of Hg

(B) 1.70 atm

(C) 816.7 m.m. of Hg

(D) 1 atm

Paragraph For Q. 18 to 20

Under the same conditions of temperature and pressure the rate of diffusion of gas is inversely proportional to the square root of vapour density. Rate of diffusion is volume of gas diffused per unit time.

$$\text{rate of diffusion} \propto \frac{1}{\sqrt{\text{vapour density}}}$$

$$\text{therefore } \frac{r_1}{r_2} = \sqrt{\frac{d_2}{d_1}} = \sqrt{\frac{2\times d_2}{2\times d_1}} = \sqrt{\frac{M_2}{M_1}}$$

Where M_1 and M_2 are molecular masses of gases. If the pressures are not same then $\frac{r_1}{r_2} = \frac{p_1}{p_2}\sqrt{\frac{M_2}{M_1}}$.

Q18. Hydrogen gas diffuses four times as rapidly as a mixture of C_2H_4 and CO_2 the molar ratio of C_2H_4 to CO_2 in the mixture is -

(A) 1 : 1 (B) 2 : 1

(C) 3 : 1 (D) 3 : 2

Q19. Two container A and B have the same volume container A contains 5 moles of O_2 gas container B contains 3 moles of He and 2 moles of N_2. Both the containers have very small orifices of same area through which the gases leak out. The rate of effusion of O_2 with that of He gas mixture will be –

(A) 0.48 (B) 0.52

(C) 0.75 (D) 0.92

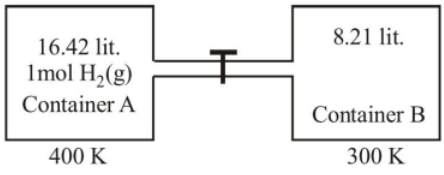
Q20. The composition of the equilibrium mixture for equilibrium Cl_2 □ 2Cl at 1470 K may be determined by the rate of diffusion of a mixture through a pin hole. It was found that at 1470 K the degree of dissociation of Cl_2 is 0.14 then the ratio of rate of effusion of equilibrium mixture to that of krypton (atomic weight = 83.8 under similar conditions will be).

(A) 1.18 (B) 3.20

(C) 5.12 (D) 4.35

Paragraph For Q. 21 to 23

A container containing $H_2(g)$ was connected to another empty container as shown in diagram and maintained at given temperature.



Q21. The final pressure in each container is -

(A) 2 atm. (B) $\frac{3}{5}$ atm

(C) $\frac{6}{5}$ atm (D) $\frac{1}{6}$ atm

Q22. The number of moles of H_2 in container B is –

(A) $\frac{3}{5}$ (B) $\frac{2}{5}$

(C) $\frac{1}{2}$ (D) $\frac{4}{7}$

Q23. The ratio of total number of collisions in container A to container B is –

(A) $\frac{3\sqrt{3}}{4}$ (B) $\frac{9}{16}$

(C) $\frac{3\sqrt{3}}{8}$ (D) None of these

Paragraph For Q. 24 to 26

The pressure of the gas is the force exerted by the gas per unit area of the walls of the container. The pressure of gases is measured by a device known as manometer. Two types of manometers, open end manometer and closed end manometer, are commonly used to measure gaseous pressure.

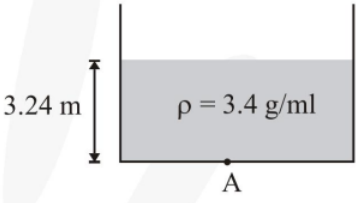
Pressure of one atmosphere (1 atm) is defined as the pressure that can support a column of mercury of 76 cm height at 0°C (density of mercury = 13.5951 g cm^{-3}) and at standard gravity = 980.665 cm s^{-2}. One atmosphere is also referred to as 760 torr.

1 atm = 76.0 cm of mercury = 760 mm of mercury = 760 torr

1 atm = 101.325×10^3 N m^{-2} = 101.325 kPa

An older unit of pressure is 'bar' which is equal to 10^5 Pa.

Q24. At point A calculate pressure due to column of liquid in atm.



(A) 3 atm (B) 2 atm

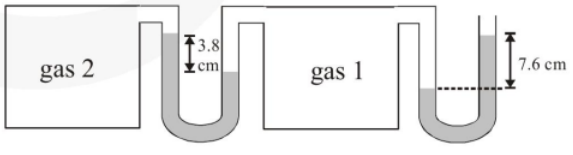
(C) 1.066 atm (D) .81 atm

Q.25 The total pressure at point A is equal to x m height of water column then x is:

(A) 10.33 m (B) 21.33 m

(C) 2.133 m (D) 11.02 m

Q26.



In above arrangement calculate pressure of gas 1 and gas 2 in atm.

(A) 1.1, 1.05 atm

(B) 1.2, 1.1 atm

(C) 1, 1.1 atm

(D) 1.05, 1.1 atm

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q27. Assume the gas to be ideal and match the following :

Column-I	**Column-II**
(A) K.E. of 4 gram of Helium molecule.	(P) $\frac{5}{2}$R
(B) Increase in the energy of 32 gram of oxygen molecule on raising temperature by 1°C.	(Q) 3 R
(C) Increase in energy of 1 mole of NO_2 on raising temperature by $\rightarrow$ 1°C at constant volume.	(R) 4 R
(D) Increases in energy of 1 mole of NO_2 on raising temperature by 1°C at constant pressure.	(S) 1.5 RT

Q28. Match the column

Column I	**Column II**
(A) H_2 gas at low P	(P) Exhibit positive deviation from ideal behavior
(B) CO_2 gas at high P	(Q) Size factor dominates the attractive forces
(C) SO_2 gas at high T and low P	(R) Volume Correction can be neglected
(D) CH_4 gas at low P	(S) Exhibit ideal behavior

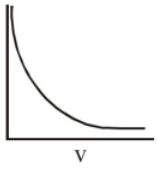
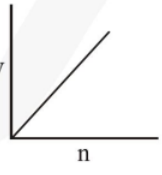
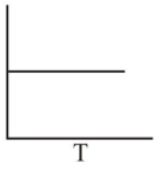
Q29.

Column-I	**Column-II**
(A) For an ideal gas of definite mass, PV	(P) Depends on temperature
(B) Saturation vapour Pressure of water	(Q) Applicable in a closed container
(C) Rate of effusion	(R) Depends on the number of gas molecules colliding with the walls of the container in unit time
(D) Pressure of an ideal gas	(S) Constant at definite Temperature

TRIPLE MATCH TYPE

*This section contains ONE table having 3 columns and 4 rows. Based on the table each question has FOUR options [A], [B], [C] AND [D]. **ONLY ONE** of these four is correct.*

For the following molecules in Column - 1, Match the correct order with Column - 2 & Column - 3 according to the question asked for constant mass of gas.

	Column - 1	**Column - 2**	**Column - 3**
I	Boyle's Law	(a) PV = Constant	(i) 
II	Charles Law	(b) $\frac{V}{T}$ = Constant	(ii) 
III	Gaylussac Law	(c) $\frac{P}{T}$ = Constant	(iii) 
IV	Avagadro Law	(d) $\frac{V}{n}$ = Constant	(iv) 

Q30. Which of the following relation is correct?

(A) I, b, (ii) (B) II, a, (iii)

(C) III, c, (ii) (D) (IV), (d), (iv)

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q31. **Statement 1 :** For H_2 and He, $Z > 1$ at all ranges of pressure.

Statement 2 : Inversion temperature is very low for H_2 and He.

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q32. At 20°C, two balloons both have equal volume and porosity are filled to a pressure of 2 atm. One with 14 kg N_2 and other with 1 kg of H_2. The N_2 balloon leaks to a pressure of 0.5 atm in 1 hr. How long (in minute) (.....) will it take for H_2 balloon to reach a pressure of 0.5 atm.

Q33. A glass bulb containing a fine orifice was filled with oxygen gas at pressure of 2 atm. It was seen in 50 mins 0.1gm of oxygen gas was effused out. Now the glass bulb is completely evacuated and filled with a gas mixture containing oxygen and another gas (x) (mol wt = 162) in the mole ratio of 3:2 at the total pressure of 10 atm. In 100 mins how many mg of x will be effused out of bulb in nearest possible integers ?

Q34. The mean free path in CO_2 at 27°C and a pressure of 10^{-9} bar having molecular diameter of 500 pm is 3.798×10^{x}. The value of x is.

Q35. The critical compressibility factor Z_C for O_2 is 0.308 and its critical pressure and critical volume are 50.1 atm and 7.8×10^{-2} litre mol^{-1} respectively. The Boyle temperature of O_2 is -

Answer Key

Ex-1

1	2	3	4	5	6	7	8	9	10
B	C	D	B	C	B	A	B	B	B
11	12	13	14	15	16	17	18	19	20
D	B	A	A	C	A	C	A	C	A
21	22	23	24	25	26	27	28	29	30
B	B	C	B	C	C	D	D	B	B
31	32	33	34	35	36	37	38	39	40
A	C	B	A	A	C	B	A	D	A
41	42	43	44	45	46	47	48	49	50
C	B	A	A	A	B	C	C	B	C
51	52	53	54	55	56	57	58	59	60
B	D	B	A	D	B	B	B	B	A
61	62	63							
D	D	A							

Ex-1A

1. A, B, D	**2.** B, C	**3.** A, B	**4.** B, C, D	**5.** B, D
6. A, B, C	**7.** A, B, D	**8.** A, B	**9.** A, B, D	**10.** A, D
11. A, C ,D	**12.** B, C	**13.** A, B, C	**14.** A, B, C, D	**15.** A, B, C
16. A, B, D	**17.** A,C	**18.** A, B	**19.** A, B, C	**20.** A, B
21. A, B, C	**22.** A, D	**23.** A, B, C	**24.** B, C, D	**25.** C, D
26. A, B, D	**27.** A, B	**28.** A, B, C	**29.** A, B	**30.** A

31. C **32.** A **33.** B **34.** A **35.** C

36. A **37.** D **38.** D **39.** C

40. A → Q,S; B → S ; C → Q; D → P

41. A → S; B → P ; C → Q; D → R

42. A → S; B → R; C → P,Q,R,S; D → P,Q

43. A → P, S; B → R; C → P, Q; D → R

44. A → Q; B → S; C → P,R; D → P,R

45. A → Q; B → S; C → P; D → R

46. B **47.** B **48.** 5 **49.** 2 **50.** 10,000 K

51. 120K **52.** 2 **53.** 4 **54.** 434.16 **55.** 0.60

56. 30

Ex-2

1	2	3	4	5	6	7	8	9	10
B	C	C	C	D	C	C	A	A	B
11	12	13	14	15	16	17	18	19	20
A	C	D	B	B	B	C	B	D	D
21	22	23	24	25	26	27	28		
C	A	D	C	A	C	B	B		

Ex-2A

1. A, B, D **2.** A, B, C, D **3.** A, B, C **4.** A, C **5.** B, C

6. D **7.** A, B, D **8.** A, C, D **9.** A, D **10.** A, C, D

11. A, C, D **12.** A, B, D **13.** A, B, C **14.** A **15.** A

16. B **17.** C **18.** C **19.** B **20.** A

21. C **22.** B **23.** C **24.** C **25.** B

26. A

27. A → P, Q, R, B → P, C → R, S D → R

28. A → P,Q,S, B → P,Q,S, C → P,R, D → P,Q,R

29. A → S, B → P, C → Q, D → R **30.** C

31. D **32.** 16.04 **33.** 900 **34.** 3 **35.** 521.57